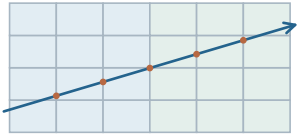
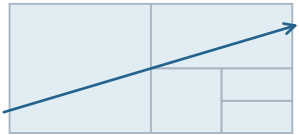
a DDA



单元穿越

索引 (i, j, k)；区间 $[t_{\text{in}}, t_{\text{out}}]$

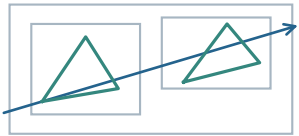
b 八叉树



叶单元定位

编号、尺寸、材料

c BVH



候选面筛选与求交

表面编号、交点参数